AW Farms Inventory — Frontend Design Spec (Phase 1)

Oct 27, 2026 · @Moosa

Scope of phase 1

Phase 1 builds the complete frontend, every screen for the Farm User and the Accounts Admin, running on mock data, so the client can click through the real product before any backend exists.

This spec follows the requirements document (AW Farms Inventory System — Requirements). Its terms, rules and decisions apply here: the same three slips (IGP, OGP, SIS), no review step, quantities only, 11 farms, and bilingual English and Urdu.

In phase 1

- Every screen in the Screen list, responsive for phone and desktop.
- All three slips working end to end in the browser: fill, validate, save, view, print.
- Stock balances that change when a slip is saved, calculated in the browser from mock data, so the demo behaves like the real thing.
- Transfers between two mock farms (OGP at one farm, waiting IGP at the other).
- Print layouts and PDF share for the three slips.
- Switching between the Farm User and Accounts Admin roles for the demo.
- A typed data layer (a mock API) whose shapes the backend will later implement unchanged.

Not in phase 1

- A real backend, database, login or permissions. Login is a mock role picker.
- Real item data. The mock uses about 60 items taken from the client's workbook and the slips.
- Excel export. The buttons are shown but not wired up.
- Data migration.

Reference: what they use today

The new screens must feel like the paper slips the farms already fill in, not like QuickBooks. These snapshots are the source for field names, column order and the pain points the design fixes. The face in the screen recording has been blurred.

The paper slips (the layout to follow)

Inward Gate Pass: header block (No., Sr. No., Time, Date, Bilty No., Truck No., Driver Name, M/s, Demand No., D. Challan No.), then a lines table with Item, Unit, Qty As Per Delivery Challan, Qty Actual Received.

SUTLEJ FEEDS LAYER

IN WARD GATE PASS

~~Jalal Pura~~ OS Broiler

No. 2048

Sr. No. _____ Time: _____ Date: 17/9/26

Bilty No 3874 Truck No. 9701 Driver Name: امین

Received the following material form M/s AW

Demand No. _____ Vide. D. Challan No. _____ Dated: _____

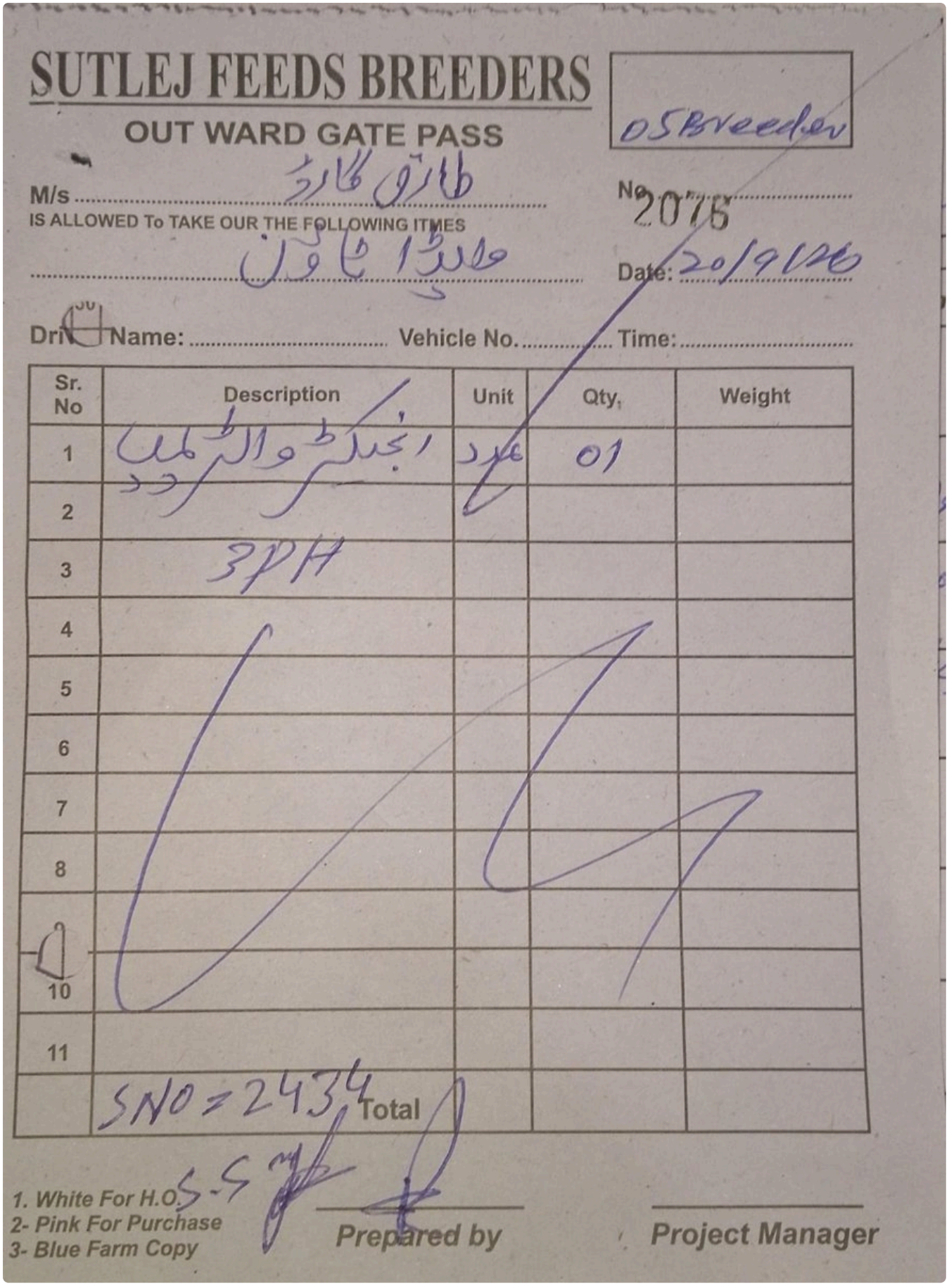
Sr. No	Item	Unit	Qty. As Per Delivery Challan	Qty. Actual Received
1	بوری	نگ	04	
2				
3	وزن		(97.550 Kg)	
4				
5				
6				
7				
8				
9	S.No = 2418			
~~10~~				
	Total			

Distribution:
1. White For H.O.
2- Pink For Purchase
3- Blue Farm Copy

Prepared by

Project Manager

Outward Gate Pass: M/s and "is allowed to take out the following items", Driver Name, Vehicle No., Time, then Description, Unit, Qty, Weight. Footer: Prepared by, Project Manager.



SUTLEJ FEEDS BREEDERS

OSBreeder

OUT WARD GATE PASS

M/s طارق گارڈ

No 2076

IS ALLOWED To TAKE OUR THE FOLLOWING ITMES

طارق گارڈ

Date: 20/9/26

Dri Name: ..... Vehicle No. ..... Time: .....

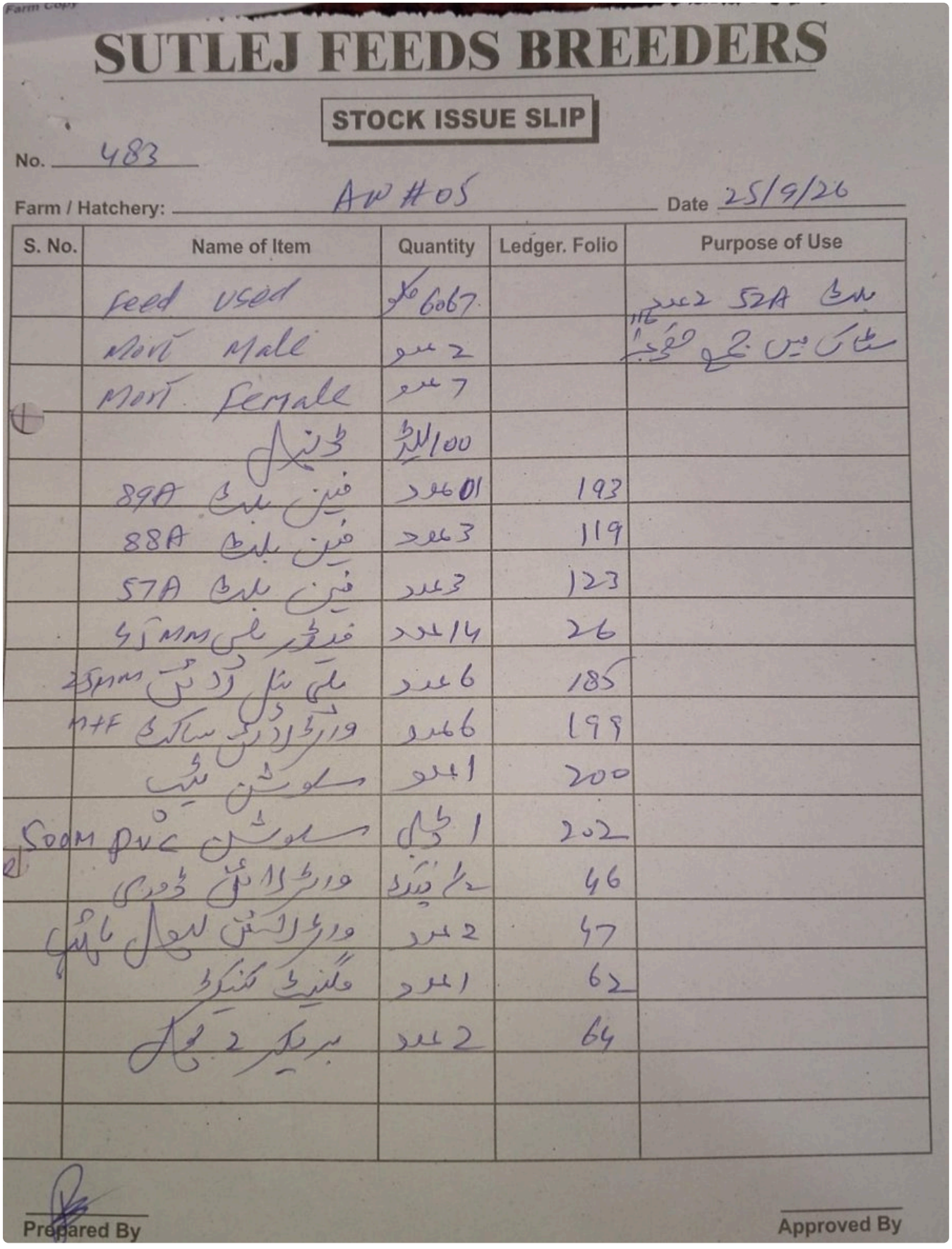
Sr. No	Description	Unit	Qty,	Weight
1	انجکشن والا ٹکلی	عدد	01	
2	"			
3	3PH			
4				
5				
6				
7				
8				
9				
10				
11				
	SNO = 2434 Total			

1. White For H.O.
2- Pink For Purchase
3- Blue Farm Copy

Prepared by

Project Manager

Stock Issue Slip: Farm / Hatchery and Date, then S. No., Name of Item, Quantity, Ledger Folio, Purpose of Use. Feed used, Mort Male and Mort Female are written on its first lines. Footer: Prepared By, Approved By.



SUTLEJ FEEDS BREEDERS

STOCK ISSUE SLIP

No. 483

Farm / Hatchery: AW #05 Date 25/9/26

S. No.	Name of Item	Quantity	Ledger. Folio	Purpose of Use
	Feed used	6067 کلو		52A بلیٹ 2 عدد
	Mort Male	2 عدد		سٹاک میں نکلے مرغے
	Mort Female	7 عدد		
	ڈنڈا	1 لیڈ/100		
	89A فیڈر پلیٹ	01 عدد	193	
	88A فیڈر پلیٹ	3 عدد	119	
	57A فیڈر پلیٹ	3 عدد	123	
	فیڈر پلی 45mm	14 عدد	26	
	25mm پلی نپل آدھ انچ	6 عدد	185	
	1HF واٹر ڈراپر سائیکی	6 عدد	199	
	سکوشن پائپ	1 عدد	200	
	50mm PVC سکوشن	1 کلو	202	
	واٹر لائن ڈوری	1 پیکٹ	46	
	واٹر لائن لیول ٹائپ	2 عدد	47	
	فلنجڈ نپل	1 عدد	62	
	بریکر 2 انچ	2 عدد	64	

Prepared By

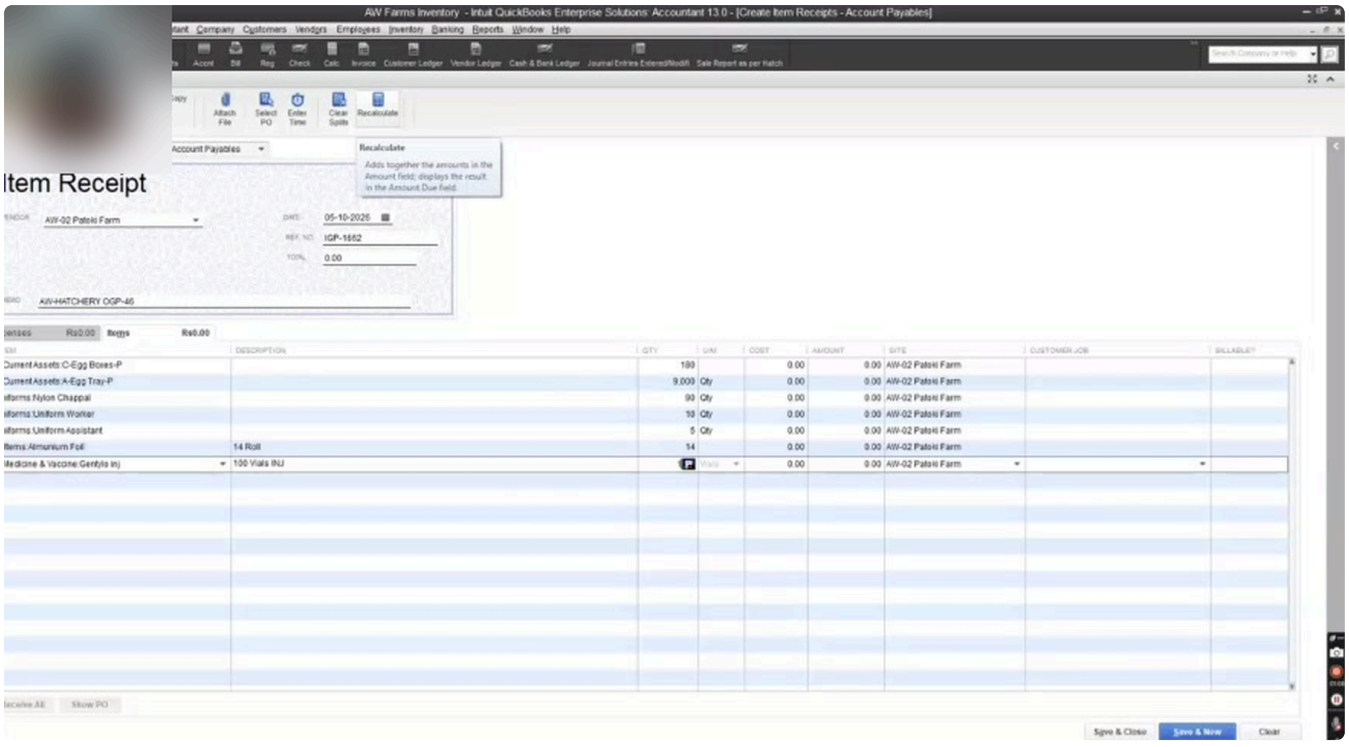
Approved By

Design notes from the slips

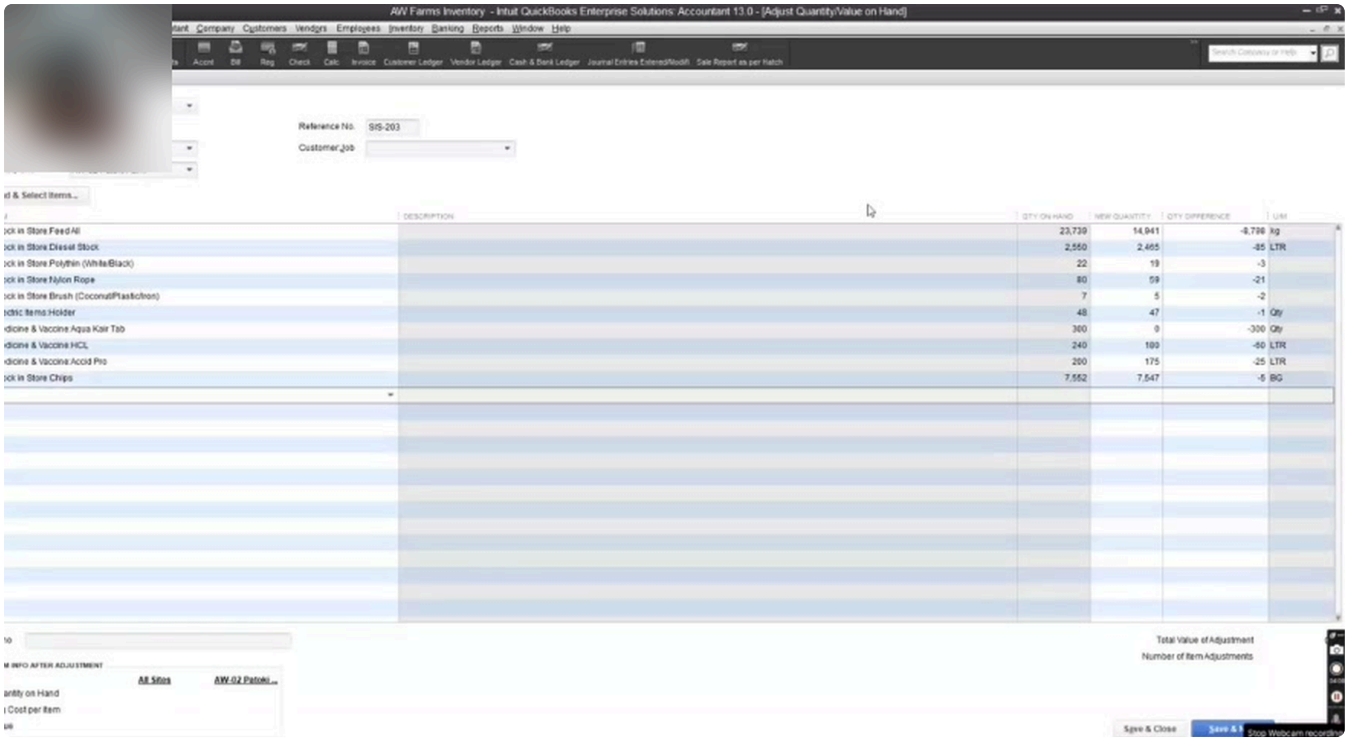
- Items are written mostly in Urdu, with codes in English ("A فین بیلٹ 89"). Search must accept both, mixed.
- Units are written as عدد (pieces), لیٹر, کلو, گڈی. The app shows the unit next to each quantity in both languages.
- Many rows are left empty, so the app adds rows one at a time instead of showing a fixed grid.
- Blank fields (Qty Actual Received, Approved By) show the paper has no way to force a field. The app marks required fields and checks them.

The QuickBooks screens (what the accounts person does today)

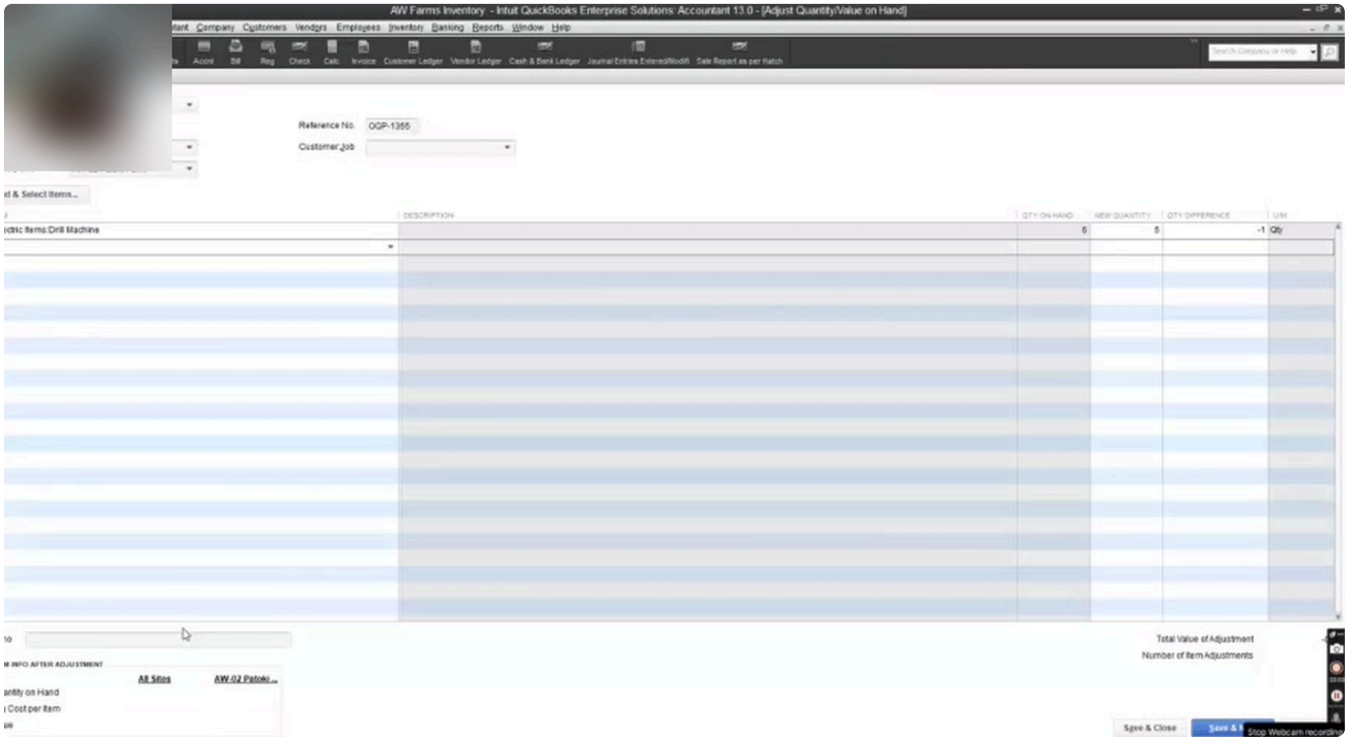
Item Receipt, used for an IGP. The farm is picked as the "vendor". The reference is the IGP number and the memo holds the sender's OGP number. Cost and Amount are always 0.00.



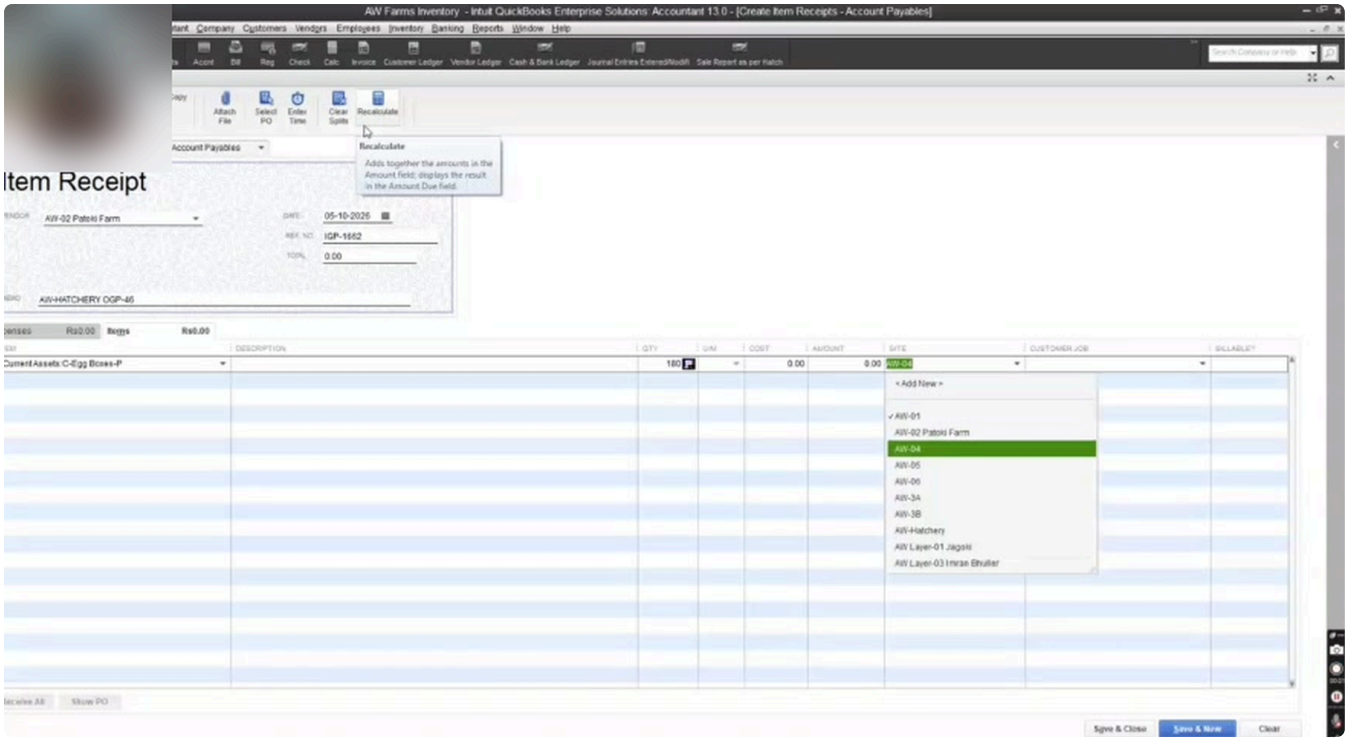
Adjust Quantity on Hand, used for an SIS. He types the **New Quantity** for each item, and QuickBooks works out the difference. Our SIS asks for the quantity used instead.



Adjust Quantity on Hand, used for an OGP. The drill machine goes 6 → 5 and the destination farm is only in the memo, so the transfer is never linked.

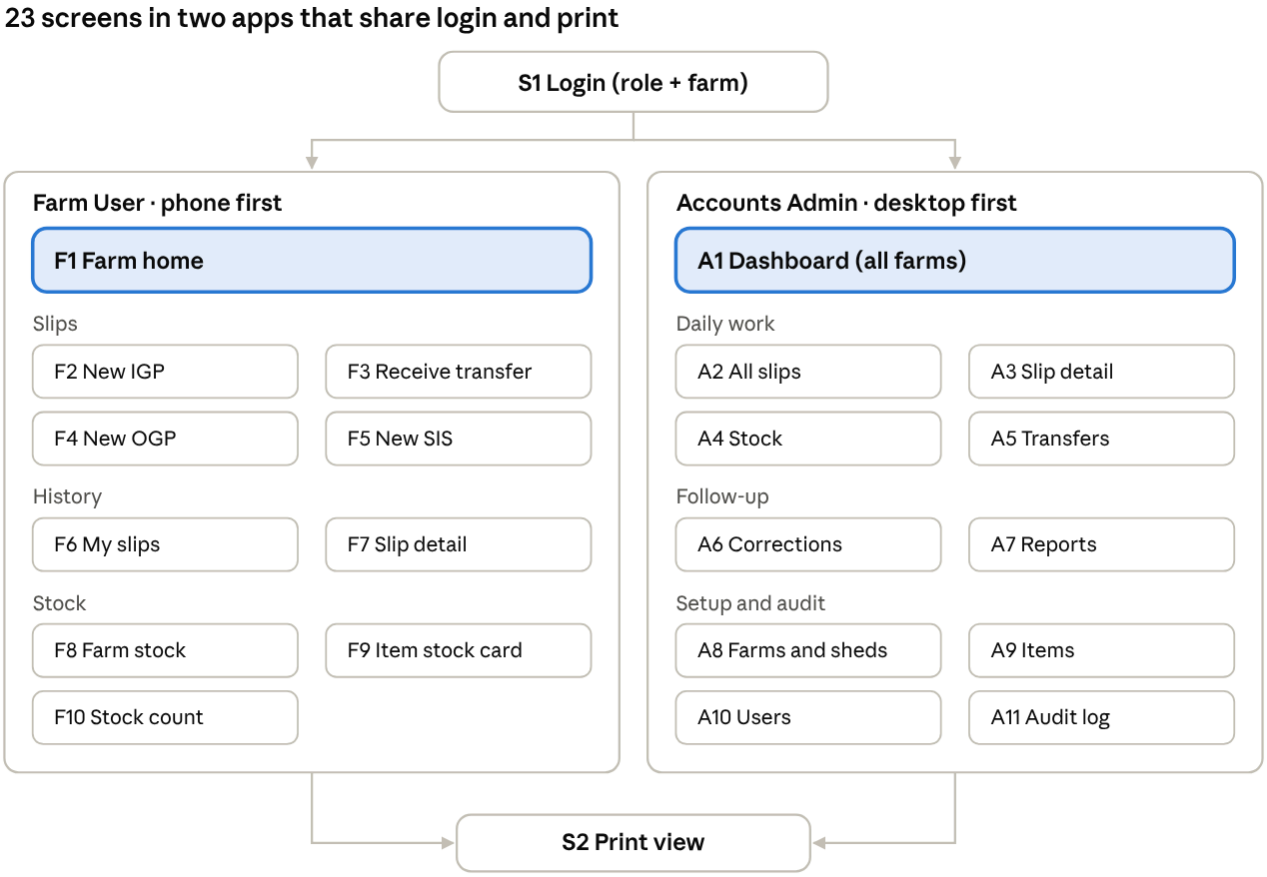


Site list = the farms. These names and codes are used as-is in the mock data.



Sitemap

One web app with two areas behind one login: the Farm User area, built for a phone, and the Accounts Admin area, built for a desktop. Both open the same print view.



sitemap · 23 screens

The Farm User reaches every slip in one tap from F1. The Accounts Admin reaches every list in one click from the sidebar.

Screen list

There are 23 screens: 10 for the Farm User (phone first), 11 for the Accounts Admin (desktop first) and 2 shared. P1 screens are needed for the first client demo, and P2 screens follow.

ID	Screen	Route	Who	Purpose	Priority
S1	Login	/login	Both	Phase 1: pick a role and a farm (mock)	P1
S2	Print view	/print/:slipId	Both	Paper layout of any slip, for printing or PDF	P1
F1	Farm home	/farm	Farm User	Three slip buttons, IGPs waiting, today's slips	P1
F2	New IGP	/farm/igp/new	Farm User	Inward Gate Pass from a supplier or other source	P1

ID	Screen	Route	Who	Purpose	Priority
F3	Receive transfer	/farm/igp/receive/:transferId	Farm User	IGP filled from the sender's OGP. Only Qty Actual Received is entered	P1
F4	New OGP	/farm/ogp/new	Farm User	Outward Gate Pass to a farm or outside party	P1
F5	New SIS	/farm/sis/new	Farm User	Stock Issue Slip with the Feed & Mortality section	P1
F6	My slips	/farm/slips	Farm User	All slips of this farm, search by No., date, item	P1
F7	Slip detail	/farm/slips/:id	Farm User	Read-only slip, print / share, request correction	P1
F8	Farm stock	/farm/stock	Farm User	Every item with quantity on hand	P1
F9	Item stock card	/farm/stock/:itemId	Farm User	Movements of one item with running balance	P2
F10	Stock count	/farm/stock-count/new	Farm User	Enter a physical count, see differences, save with reason	P2
A1	Dashboard	/admin	Accounts Admin	Today's slips, in transit, shortfalls, correction requests, farms missing SIS	P1
A2	All slips	/admin/slips	Accounts Admin	Every slip from every farm, filters, open detail	P1
A3	Slip detail (admin)	/admin/slips/:id	Accounts Admin	As F7, plus Correct and Cancel	P1
A4	Stock	/admin/stock	Accounts Admin	Farm × item matrix with All Farms total, drill into a farm	P1
A5	Transfers	/admin/transfers	Accounts Admin	Every OGP between farms with status and shortfall	P1
A6	Correction requests	/admin/corrections	Accounts Admin	Requests from farms: approve with correction, or reject	P2
A7	Reports	/admin/reports/:type	Accounts Admin	The 8 reports from the requirements	P2
A8	Farms & sheds	/admin/setup/farms	Accounts Admin	Add or edit farms and their sheds	P2
A9	Items	/admin/setup/items	Accounts Admin	Item list with category, unit, Urdu name, aliases	P2
A10	Users	/admin/setup/users	Accounts Admin	One Farm User per farm, plus the suggested view-only users	P2

ID	Screen	Route	Who	Purpose	Priority
A11	Audit log	/admin/audit	Accounts Admin	Who did what, when, old and new values, reason	P2

If the client accepts the suggested view-only login for Head Office and Purchase, it reuses A1–A5 and A7 with every action button hidden. No new screens are needed.

Farm User screens

Every Farm User screen is designed for a 360 px wide Android phone first. Each slip is one scrolling page that reads top to bottom like the paper: header, lines, footer, Save.

F1 Farm home

- Top bar: farm name (AW-02 Patoki Farm), user name, language order toggle.
- Three large buttons, full width, each with icon, English and Urdu: **Inward Gate Pass / آمد گیٹ پاس**, **Outward Gate Pass / روانگی گیٹ پاس**, **Stock Issue Slip / اشیاء جاری پرچی**.
- "Waiting to receive" card: one row per incoming transfer (from farm, OGP No., date, number of items). Tapping it opens F3. Hidden when empty.
- "Today's slips" list: No., type chip, time, first item + "and 4 more".
- Bottom navigation: Home, Slips, Stock.

Slip header and footer (F2–F5)

- Slip title in English and Urdu, with the slip No. shown as "New" until saved.
- Farm / Hatchery, Date and Time fill in automatically and are shown as read-only text. Time can be adjusted (tap "change") when a slip is entered later the same day.
- Footer: "Prepared by" with the user's name, and a sticky Save button at the bottom of the screen.

F2 New IGP

Field	Control	Required	Rules
Received from M/s	Segmented: Supplier / Other. Then a searchable list of past suppliers with "add new"	Yes	Farms are not offered here. Transfers come in through F3
Bilty No.	Text	No	

Field	Control	Required	Rules
Truck No.	Text	No	
Driver Name	Text, Urdu allowed	No	
Demand No.	Text	No	
D. Challan No. and Dated	Text + date	No	
Lines: Item	Item search (C1)	Yes, at least 1 line	
Lines: Unit	Shown from item	—	Not editable
Lines: Qty As Per Delivery Challan	Quantity input (C2)	Yes	> 0
Lines: Qty Actual Received	Quantity input (C2)	Yes	⩾ 0. If less than challan, the row turns amber and shows "short by 3"
Lines: Weight	Number + kg	No	

F3 Receive transfer

- Opened from "Waiting to receive". The header shows From farm, OGP No., Date sent, Driver, Vehicle, all read-only.
- Lines are pre-filled from the OGP: Item, Unit and Qty As Per Delivery Challan (= qty sent) are read-only.
- Only **Qty Actual Received** is entered. A "Received all" button fills every row with the sent quantity.
- More than sent is blocked: "Cannot receive more than 9,000. Extra items need a separate IGP."

F4 New OGP

Field	Control	Required	Rules
To	Segmented: Another farm / Outside party	Yes	
Farm	Farm picker (the other 10 farms)	If another farm	

Field	Control	Required	Rules
M/s	Text, Urdu allowed	If outside party	
Reason	Chips: Repair / Return to supplier / Other	If outside party	
Driver Name, Vehicle No.	Text	No	
Lines: Description (item)	Item search (C1), shows "in stock: 6" under the name	Yes	
Lines: Unit	From item	—	
Lines: Qty	Quantity input (C2)	Yes	> 0 and ⩽ in stock, else red "Only 6 in stock"
Lines: Weight	Number + kg	No	

After saving, the screen goes straight to the print view (S2), because the driver needs the pass.

F5 New SIS

Two sections, in the order of the paper slip:

1. **Feed & Mortality.** One row per shed of this farm, all sheds listed. Columns: Shed, Feed (item picker limited to feed items, default the last one used), Feed used (KG), Mort Male, Mort Female. Empty rows are skipped on save. A total row sits under it.
2. **Items.** Add rows one at a time. Columns: S. No. (automatic), Name of Item (C1, showing in stock), Quantity used (C2, ⩽ in stock), Ledger Folio (the item number, read-only, tappable), Purpose of Use (text, Urdu allowed, optional shed chip).

Before saving, a summary sheet lists every line as "Diesel: 2,550 → 2,465 LTR", so the user sees the new balances they used to calculate by hand. They then confirm.

F6–F10

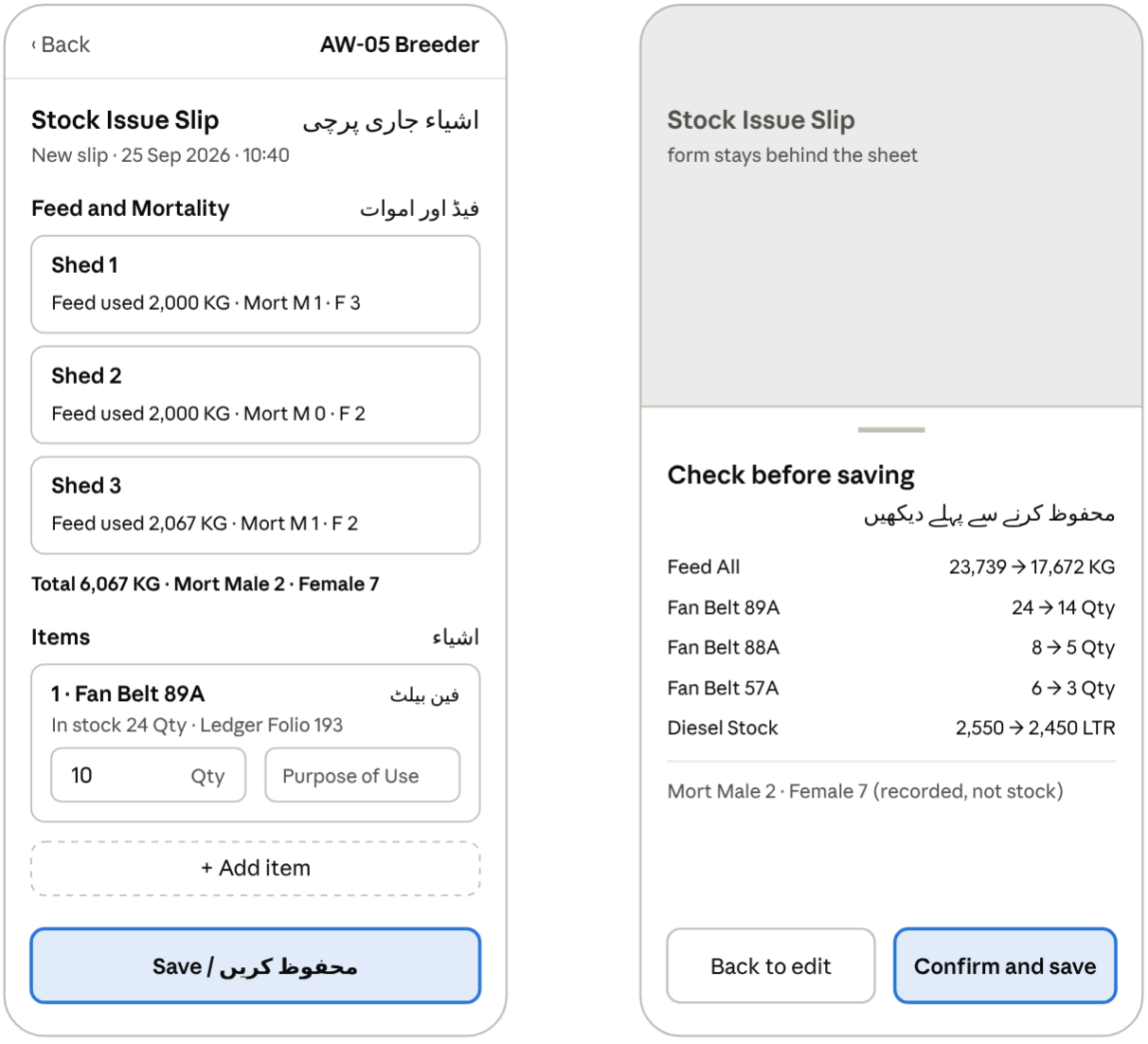
- **F6 My slips.** A search box (No., item), filter chips (IGP / OGP / SIS, date), and cards with No., type, date and line count.
- **F7 Slip detail.** The slip laid out like the paper, read-only, with Print, Share PDF and Request correction. A corrected or cancelled slip shows a banner linking to the correction.

- **F8 Farm stock.** A search box, category filter, and a list of item name (English and Urdu), quantity and unit. Items at 0 are greyed out at the bottom.
- **F9 Item stock card.** The item header with current balance, then a list of date, slip No., in, out and running balance.
- **F10 Stock count.** Pick a category or all items. Each row shows system qty and a counted-qty input, and the difference is highlighted. A reason is required before saving.

F5 wireframe

The SIS keeps the paper order and shows new balances before saving

Wireframe at 360 px · sample values from paper SIS No. 483



F5 New SIS **Summary sheet (C9)**

F5 wireframe · form and confirm sheet

The form follows the paper slip from top to bottom. The confirm sheet does the arithmetic the accounts person does by hand in QuickBooks today.

Accounts Admin screens

The Accounts Admin works on a desktop, usually 1366 × 768. The layout has a left sidebar, a farm filter in the top bar (All farms or one farm) that applies to every screen, and dense tables.

Sidebar: Dashboard, Slips, Stock, Transfers, Corrections, Reports, Setup (Farms & sheds, Items, Users), Audit log. It collapses to icons below 1024 px.

A1 Dashboard

- **Top row, five count tiles, each clickable:** Slips today, In transit, In transit > 2 days, Open shortfalls, Correction requests.
- **Farms today:** a compact table with one row per farm (11): IGP count, OGP count, SIS saved (tick or "missing" in amber), last slip time.
- **Latest slips:** the 20 newest across all farms, live: time, farm, No., type, first item.

A2 All slips and A3 Slip detail

- **A2 table columns:** No., Type, Farm, Date, From / To, Lines, Prepared by, Status (Saved, Corrected, Cancelled).
- **A2 filters:** type, farm, date range, item, status. Search by No.
- **A3** shows the slip like F7, plus **Correct** (opens the slip lines with editable quantities and a required reason, then shows the before / after difference) and **Cancel** (asks for a reason and warns if it is a received transfer, which cannot be cancelled).
- Corrections and the original are linked in both directions.

A4 Stock

- **All farms view:** a matrix with items as rows and the 11 farms as columns, plus an All Farms total. The first column (item) and the header stay fixed while scrolling. Category filter and item search.
- **One farm view:** the same list as F8, with a link to each item's stock card.
- An "In transit" column shows the quantity sent but not yet received.

A5 Transfers

- **Table columns:** OGP No., From, To, Sent, Received, Status (In transit, Received, Short), Shortfall, Age in days.
- Opening a row shows the sent and received lines side by side, with a **Resolve** action for shortfalls: write off with reason, or keep open.

A6 Correction requests

- A list of requests: farm, slip No., line, requested quantity, reason, date.
- **Approve** opens A3′s Correct flow pre-filled. **Reject** asks for a note that the farm sees on F7.

A7 Reports

- One page per report (Stock on hand, Item stock card, Consumption, Feed & mortality, Gate register, Transfers, Out for repair, Corrections).
- Each has the same filter bar (farm, date range, category or item), a table, and Print / PDF / Excel buttons (Excel not wired in phase 1).
- Feed & mortality adds a small line chart of daily mortality per shed.

A8–A11 Setup and audit

- **A8 Farms & sheds:** farm code, name, type, Farm User, sheds list (add, rename, deactivate). Farms and sheds are never deleted, only deactivated.
- **A9 Items:** table of name, Urdu name, aliases, category, unit, active. An edit drawer opens on the right.
- **A10 Users:** name, phone, role, farm, active. Phase 1 shows the screen with mock data only.
- **A11 Audit log:** time, user, action, slip or record, old → new, reason. Filter by user, farm, date.

Shared components

Twelve components carry almost every screen. Building them first, in isolation with Storybook, makes the screens quick to assemble.

ID	Component	What it does	Key details
C1	Item search	Finds an item by English name, Urdu name, alias or code, and returns item + unit	Matches mixed input ("belt 89A", "فین بیلٹ"), ignores spelling of Urdu letters like ي/ی and ك/ک, shows English, Urdu and "in stock" in each result, recent items first. Full-screen sheet on phone, dropdown on desktop

ID	Component	What it does	Key details
C2	Quantity input	Number entry with the unit fixed beside it	Numeric keypad on phone, thousands separators (9,000), decimals only for KG and LTR, shows the error below the field
C3	Slip lines	The editable table of lines on every slip	Card per line on phone, table row on desktop. Add line, remove line, auto S. No.
C4	Slip header	Title, No., Farm, Date, Time	The same on every slip, bilingual title
C5	Bilingual label	English and Urdu text for one label	Both texts in one component. The order follows the user's setting
C6	Type chip	IGP, OGP, SIS	Each type has its own colour and icon, used everywhere a slip appears
C7	Status chip	Saved, Corrected, Cancelled, In transit, Received, Short	Colour + word, never colour alone
C8	Farm picker	Pick one of the 11 farms, or All farms (admin)	Shows code + name
C9	Summary sheet	The confirm step before saving	Lists every line as "before → after"
C10	Data table	Admin tables	Sort, filter bar, fixed header, fixed first column, pagination of 50
C11	Filter bar	Farm, date range, type, item	The same filters on every admin list and report
C12	Print frame	Wraps a slip in its paper layout	A4 and A5 page sizes, print CSS

Visual design

The look is plain, calm and high-contrast. It has to read in sunlight on a cheap phone and make the three slip types impossible to mix up.

Bilingual text

- Every label, button, title and menu item shows English and Urdu. By default English is on top and Urdu below it, at 1.15× the size. Each user can swap the order (Urdu first) in their profile.
- The page direction stays left to right. Urdu text runs right to left inside its own element (`dir="rtl"`, `lang="ur"`).
- Numbers, dates, slip numbers and item codes are always in English digits (0–9) and left to right, even inside Urdu text. The paper slips already use English digits.
- Free-text fields (Driver Name, M/s, Purpose of Use) detect the language of what is typed and set the direction automatically (`dir="auto"`).
- All strings live in one translation file per language, keyed by id. No text is hard-coded in components.

Typography

Use	English	Urdu
Font	Inter	Noto Nastaliq Urdu
Body	16 px / 24	18 px / 36 (Nastaliq needs about 2× line height)
Small (helper, table)	14 px / 20	16 px / 32
Slip title	20 px semibold	22 px
Numbers in tables	Inter with tabular figures, right-aligned	—

Nastaliq is tall and its descenders clip easily, so containers with Urdu text never have a fixed height.

Colour

Token	Use	Value
primary	Buttons, links, active nav	Deep green #1F6B4A
igp	Inward Gate Pass chip and icon	Green #2E7D32
ogp	Outward Gate Pass chip and icon	Orange #C2571A
sis	Stock Issue Slip chip and icon	Blue #1E5AA8

Token	Use	Value
warning	Shortfall, missing SIS, > 2 days in transit	Amber #B7791F on #FDF4E3
danger	Errors, not saved, over stock	Red #C62828 on #FDECEC
ink / muted	Text	#1A1D1F / #5B6166
surface / line	Background, borders	#FFFFFF, #F5F6F4 / #E1E4E0

All text meets WCAG AA contrast. Colour always comes with a word or icon. Dark mode is not needed.

Spacing, touch and layout

- 4 px spacing scale (4, 8, 12, 16, 24, 32, 48).
- Touch targets at least 48 × 48 px on the Farm User screens. The Save button is 56 px tall and sticky.
- Corners: 8 px on cards and inputs, 999 px on chips.
- Icons: Lucide, 24 px, always next to a word.

Breakpoints

Width	Layout
< 640 px	Phone: one column, bottom nav, slip lines as cards
640–1023 px	Tablet: one wide column, slip lines as a table
≥ 1024 px	Desktop: sidebar + content, admin tables full width

The Farm User screens are designed at 360 px and the Accounts Admin screens at 1366 px. Both must still work at every other width.

States and messages

Every screen designs its empty, loading and error states, not just the filled one. Messages are short, bilingual and say what to do next.

State	Where	What the user sees
Loading	Any list or slip	Skeleton rows in the shape of the content, never a blank page

State	Where	What the user sees
Empty	Lists, waiting to receive, search	One line + the next action, e.g. "No slips today. Start with the buttons above."
Saving	Save button	Button shows a spinner and is disabled, so a second tap does nothing
Saved	After any slip	Green banner "IGP AW02-IGP-1663 saved" + Print / Share / New slip. Form clears
Not saved	Connection lost while saving	Red banner "Not saved. Check internet and tap Save again." Everything typed stays
Field error	Under a field	Red text, e.g. "Only 6 in stock", "Enter Qty Actual Received"
Shortfall	IGP line, transfer	Amber row + "Short by 3". Not an error, the slip can be saved
Unsaved changes	Leaving a half-filled slip	"Leave without saving? The slip will be lost." with Stay / Leave
Corrected / Cancelled	Slip detail	Banner at the top with date, by whom, reason, and a link to the correction
Not allowed	Admin-only action, wrong farm	Hidden, not disabled, for the Farm User

The unsaved half-filled slip is kept in the browser while the tab is open, so an accidental reload does not lose it.

Print layouts

The printed slip copies the current paper slip so gate staff, drivers and Head Office recognise it at once. Only the copy-colour footer goes, because one record replaces the three copies.

- **Header:** "SUTLEJ FEEDS BREEDERS" or "SUTLEJ FEEDS LAYER" (by farm type), slip title in a box, farm code in the top-right box as on the paper, No., Date, Time.
- **Body:** the same fields and table columns, in the same order as the paper slip (see Reference). Empty rows are not printed.
- **Footer:** Prepared by (name printed), and blank signature lines for Project Manager (gate passes) or Approved By (SIS), as today.

- **QR code** at the top right, which opens the slip in the app. On an OGP to another farm, scanning it at the receiving farm opens F3 directly.
- **"Corrected" or "Cancelled" stamp** across the page when that applies.
- **Sizes:** A5 portrait by default (the paper books are about A5), A4 as an option. Black and white only, so it prints on any printer.
- **Share:** the same layout as a PDF, shared from the phone's share sheet (WhatsApp).

Urdu in print uses the same Nastaliq font, embedded in the PDF. This needs testing on a real printer early in the build.

Mock data and data shapes

The frontend talks only to a small API module. In phase 1 that module is backed by mock data kept in the browser. Later, the same functions call the real backend, and no screen changes.

Mock data set

- **Farms:** the 11 sites from QuickBooks (AW-01, AW-02 Patoki Farm, AW-04, AW-05, AW-06, AW-3A, AW-3B, AW-Hatchery, AW Layer-01 Jagoki, AW Layer-03 Imran Bhullar, Layer-04 Khan), each with 3–6 sheds.
- **Items:** about 60 real items with units and Urdu names, taken from the workbook, the recording and the slips (Feed All KG, Diesel Stock LTR, Egg Tray-P, Egg Boxes-P, Fan Belt 89A / 88A / 57A, Drill Machine, Gentylo inj, Aqua Kair Tab, HCL, Accid Pro, Nylon Rope, Chips BG …).
- **Balances:** opening stock per farm, e.g. Patoki with Feed All 23,739 KG and Diesel 2,550 LTR as in the recording.
- **Slips:** about 2 weeks of history across farms, including the real examples (IGP-1662 from the Hatchery, OGP-1366 drill machine, SIS-203), one transfer in transit, one short transfer and one corrected slip.
- **Users:** one Farm User per farm and one Accounts Admin.

Data shapes (TypeScript)

```
type Unit = 'KG' | 'LTR' | 'QTY' | 'BG' | 'ROLL' | 'VIAL';
type SlipType = 'IGP' | 'OGP' | 'SIS';
type SlipStatus = 'saved' | 'corrected' | 'cancelled';

interface Farm { id: string; code: string; name: string; type: 'breeder' |
'layer' | 'hatchery'; sheds: Shed[]; active: boolean }
interface Shed { id: string; farmId: string; name: string; active: boolean }
interface Item { id: string; number: number; name: string; nameUr: string;
aliases: string[]; category: string; unit: Unit; isFeed: boolean; active:
```

```
boolean }

interface SlipBase {
  id: string; no: string;            // AW02-IGP-1663
  type: SlipType; farmId: string;
  date: string; time: string;        // ISO date, HH:mm
  preparedBy: string; status: SlipStatus;
  correctionOf?: string; correctedBy?: string;
}

interface IGP extends SlipBase {
  type: 'IGP';
  source: { kind: 'supplier' | 'farm' | 'other'; farmId?: string; ms?: string
};
  transferId?: string; biltyNo?: string; truckNo?: string; driverName?:
string;
  demandNo?: string; challanNo?: string; challanDate?: string;
  lines: { itemId: string; qtyChallan: number; qtyReceived: number;
weightKg?: number }[];
}

interface OGP extends SlipBase {
  type: 'OGP';
  to: { kind: 'farm' | 'outside'; farmId?: string; ms?: string; reason?:
'repair' | 'return' | 'other' };
  transferId?: string; driverName?: string; vehicleNo?: string;
  lines: { itemId: string; qty: number; weightKg?: number }[];
}

interface SIS extends SlipBase {
  type: 'SIS';
  shedRows: { shedId: string; feedItemId: string; feedKg: number; mortMale:
number; mortFemale: number }[];
  lines: { itemId: string; qty: number; purpose?: string; shedId?: string }
[];
}

interface Transfer { id: string; ogpId: string; igpId?: string; fromFarmId:
string; toFarmId: string; status: 'in_transit' | 'received' | 'short';
sentAt: string; receivedAt?: string }
interface Movement { id: string; farmId: string; itemId: string; slipId:
string; date: string; qtyIn: number; qtyOut: number }
```

API functions (mocked in phase 1): `listFarms`, `listItems`, `searchItems(q)`, `getStock(farmId)`, `getStockCard(farmId, itemId)`, `listSlips(filters)`, `getSlip(id)`, `saveIGP`, `saveOGP`, `saveSIS`, `listWaitingTransfers(farmId)`,

`receiveTransfer`, `requestCorrection`, `correctSlip`, `cancelSlip`, `saveStockCount`, `getDashboard`, `getReport(type, filters)`.

Balances are always calculated from the movements. Save functions reject a line that would take stock below zero, with the same message the screen shows.

Tech stack and project structure

We suggest a TypeScript stack that a later backend can plug into without rewriting screens.

Concern	Choice	Why
Framework	Next.js (App Router) + TypeScript	Routing for 23 screens, easy deploy for client demos
Styling	Tailwind CSS with the colour and spacing tokens above	Tokens in one place, fast to build
Components	shadcn/ui (Radix)	Accessible dialogs, sheets, comboboxes to restyle
Forms	React Hook Form + Zod	The Zod schemas also hold the slip rules (required, ⩽ stock)
Data	TanStack Query over the API module	Loading, error and refetch states for free. The mock and real APIs swap behind it
Mock backend	In-memory store saved to localStorage, seeded from JSON	Demo data survives a reload. A "reset demo" button restores it
Translation	next-intl with en.json and ur.json	All bilingual labels from one place
Fonts	Inter + Noto Nastaliq Urdu (self-hosted)	No dependence on a font CDN at farms
Item search	Fuse.js with Urdu letter normalisation	Fuzzy match across English, Urdu and aliases
Print / PDF	Print CSS for printing, react-pdf for the shared PDF	Exact page layout, embedded Urdu font
Component review	Storybook	Show C1–C12 to the client before the screens
Hosting for demos	Vercel preview links	One link per milestone for the client

```
src/
  app/(farm)/farm/...        F1–F10
  app/(admin)/admin/...      A1–A11
  app/print/[slipId]/        S2
  components/                C1–C12
  features/igp|ogp|sis/      slip forms, schemas, summary
  lib/api/                   api.ts (interface), mock/ (store + seed)
  lib/stock.ts               balance from movements
  messages/en.json, ur.json
```

Deliverables and timeline

We estimate about 7 weeks for one frontend developer. The client sees something clickable at the end of week 1 and every week after.

Week	Build	Client sees
1	Design tokens, fonts, bilingual label setup, low-fidelity wireframes of F1–F5 and A1	Wireframe link to confirm layout and field order against the paper slips
2	Components C1–C12 in Storybook, mock store and seed data	Item search in English and Urdu, the slip lines on a phone
3	F1, F2, F3, F5	Fill and save an IGP and an SIS, with balances changing
4	F4, transfers end to end, F6–F8, print view (S2)	Send an OGP from one farm and receive it at another. Print and PDF
5	A1–A5 (dashboard, slips, correct and cancel, stock, transfers)	The Accounts Admin's live view of all 11 farms
6	P2 screens: F9, F10, A6–A11	Reports, stock count, setup screens
7	Urdu review with a native reader, test on a low-cost Android phone and a real printer, fixes	Final demo build for sign-off

Handed over at the end: the running demo link, the source code, Storybook, the translation files, and the API module with its types for the backend developer.

Open design questions

These change what the screens look like, so they are best asked at the week-1 wireframe review.

- [ ] **Language order:** should English or Urdu come first by default for Farm Users?
- [ ] **Urdu slip names:** we drafted آمد گیٹ پاس, روانگی گیٹ پاس, اشیاء جاری پرچی. Do the farm staff use other words?
- [ ] **Branding:** is there a logo? Are "Sutlej Feeds Breeders" and "Sutlej Feeds Layer" the right names for the print header?
- [ ] **Feed:** one feed item or several feed types in the SIS Feed & Mortality section? (open in the requirements too)

- [ ] **Mortality:** do layer farms and the hatchery use the same Mort Male / Mort Female columns?
- [ ] **Sheds:** names and number per farm, for the mock data.
- [ ] **Printers:** A5 or A4? Which farms print?
- [ ] **Phones:** which phone models do Farm Users have, so we test on the right screen size?
- [ ] **View-only login:** if accepted, does Head Office need any report the Accounts Admin doesn't?